

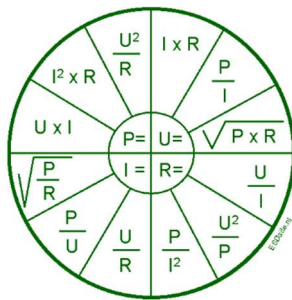
Doel:

Het onderzoeken en aanleren van hoe een led zich gedraagt ifv stroom en spanning.

En deze dan ook zelf in een grafiek opnemen.

Formules:

de eigenschappen van een serie-schakeling zijn hier van toepassing



Benodigheden:

Weerstanden:

Type: E12 merk: Velleman

Merk op dat in dit verslag de berekende waarden niet de fysieke waarden zijn daar er maar sommige typen bestaan (E12 reeks).

Men zal dan bij benadering de dichtstbij gelegen waarde nemen naar boven.

- $R1 = 120 \Omega$

Led:

Type : clear blue 5mm merk: generieke (besteld via Temu) $V_f = 2.7 \text{ V}$ $I_n = 20\text{mA}$

(<https://www.we-online.com/components/products/datasheet/151053BS04500.pdf>)



Breadboard :

Merk Velleman



OPDRACHT 2: SPANINGSDELING EN LED'S

PBA EO 1

Kyell De Windt

Mr.mees
Mr.Vereecke

Jumperkabels voor breadboards



HobbyElectronica.nl

Meetpenen van Velleman



Voedingsbron : merk Keysight

type : EDU36311A

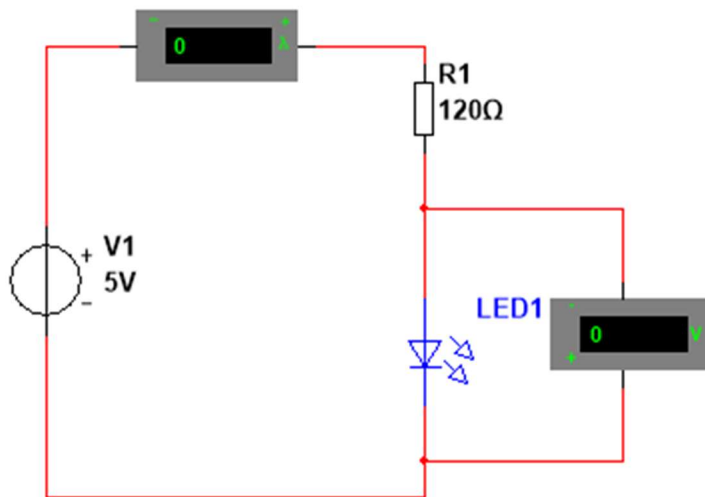


Meettoestel : merk keysight

type : EDU34450A 5 ½ digit Multimeter



Schema



$$R_{led} = \frac{U_{bron} - U_{led}}{I_{led}} = \frac{5V - 2.7V}{0,020A} = 120 \Omega$$

$$P_{LED} = U_{LED} * I_{LED} = 2.7V * 0.020A = 0.054 W$$

$$P_{R1} = U_{R1} * I_{LED} = 2.3V * 0.020A = 0.046 W$$

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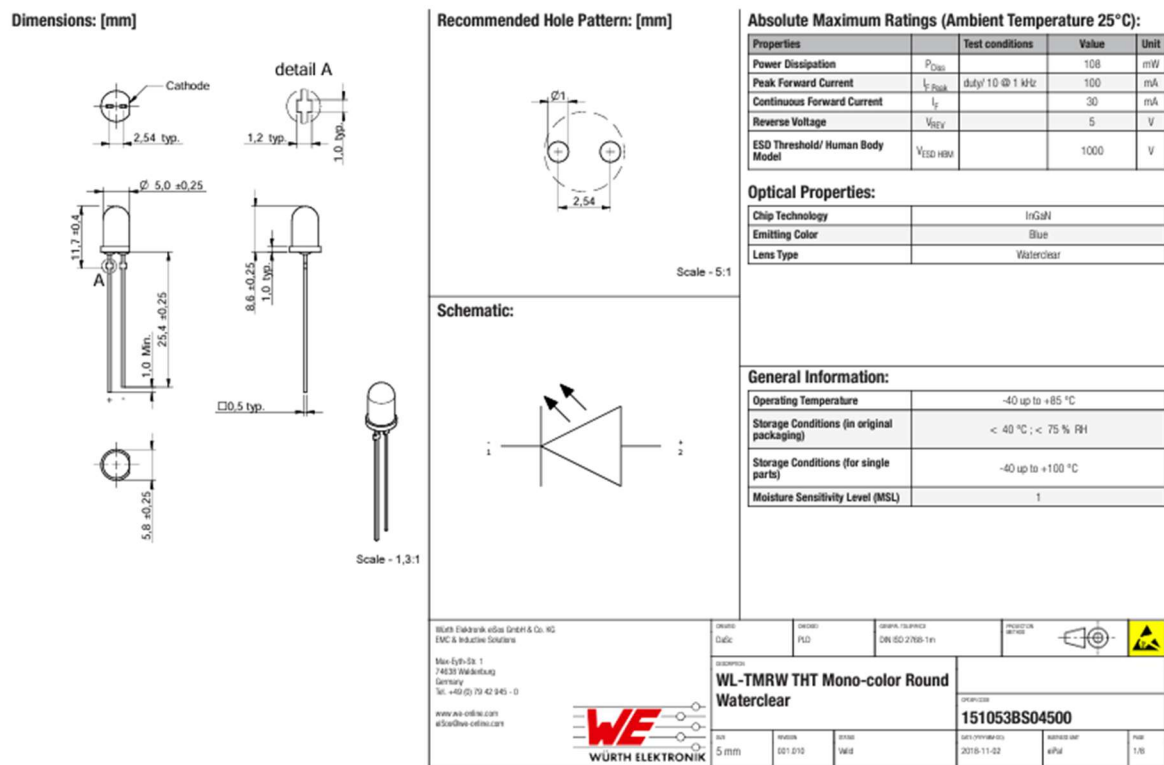
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Datasheet

Uitleg :

Een datasheet is een document(meestal pdf) van een product waarin de fabrikant alle gegevens van de gedane testen in plaatst zodoende dat de gebruiker de juiste info heeft om eventuele projecten tot een goed einde te brengen.



Op deze pagina kan men volgende dingen terug vinden:

Dimensions

de afmetingen van het product (grootte en vorm) dit is belangrijk als men bv pcb willen ontwerpen de design kunnen afstemmen op deze gegevens.

Recommended hole pattern

Dit is de afmeting die in ons geval tussen de "pootjes" van de led nodig is en dikte van de gaten (belangrijk voor montering in pcb).

Schematic

Het schematisch symbool van in ons geval de led

Absolute Maximum Ratings (Ambient Temperature 25°C)

De maximale waarden onder een omgevings-temperatuur van 25 graden celcius.

Power Dissipation (vermogens dissipatie)

De hoeveelheid vermogen die led kan verbruiken .

Peak Forward Current (piek voorwaardse stroom)

De maximale kortstondige toegelaten stroom door de led.

Continuous Forward Current (constate voorwaardse stroom)

De maximale constante toegelaten stroom door de led.

Reverse Voltage (omgekeerde spanning)

De maximale omgekeerde spanning die de led kan verdragen zonder kapot te gaan.

ESD Threshold/ Human Body Model (electro statische ontlading / menselijk lichaam model)

De maximale statische lading een led kan verdragen alvorens hij doorslaat 'kapot gaat'.

Optical Properties: (optische eigenschappen)

De visuele eigenschappen van de led

Chip Technology (chip technologie)

Het soort chip materiaal en type de led is gemaakt

Emitting Color (uitzend kleur)

Het uitgezonden kleur van de led

Lens Type (het soort lens die gebruikt is)

In ons geval is dat een waterclear versie

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General Information: (algemene info)

Operating Temperature (temp bereik voor normale werking)

De normale werkings-temperatuur van de led

Storage Conditions (gegevens rond het opslaan en bewaren van de componenten)

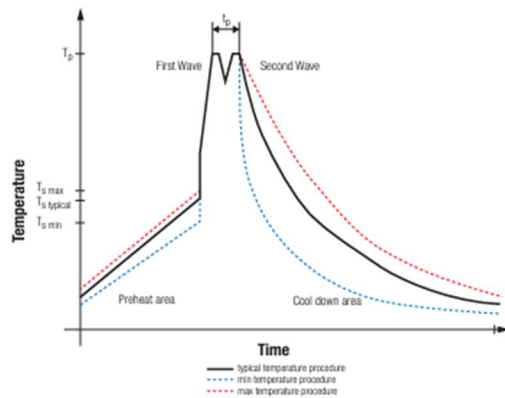
De manier van bewaren zonder dat er iets met het component kan gebeuren.

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Classification Wave Soldering Profile:



Classification Wave Soldering Profile:

Profile Feature	Pb-Free Assembly	Sn-Pb Assembly
Preheat Temperature Min	$T_{s \min}$ 100 °C	100 °C
Preheat Temperature Typical	$T_{s \text{ typical}}$ 120 °C	120 °C
Preheat Temperature Max	$T_{s \max}$ 130 °C	130 °C
Preheat Time t_p from $T_{s \min}$ to $T_{s \max}$	t_p 70 seconds	70 seconds
Ramp-up Rate	ΔT 150 °C max.	150 °C max.
Peak Temperature	T_p 245 °C - 260 °C	235 °C - 260 °C
Time of actual peak temperature	t_p 5-8 seconds max.	3-6 seconds max.
Ramp-down Rate, Min	~ 2 K/ second	~ 2 K/ second
Ramp-down Rate, Typical	~ 3.5 K/ second	~ 3.5 K/ second
Ramp-down Rate, Max	~ 5 K/ second	~ 5 K/ second
Time 25 °C to 25 °C	4 minutes	4 minutes
Applied cycles	1 cycle max.	

refer to EN61760-1:2006

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DATE: Date:	REV: P02	QUALITY STANDARD: DIN ISO 27001-TM	PRODUCTION SMT-005
DESCRIPTION: WL-TMRW THT Mono-color Round Waterclear			ORDER CODE: 1510538S04500
Lot: 5 mth	VERSION: 001.010	STATUS: VALID	DATE OF RELEASE: 2016-11-02

This electronic component has been designed and developed for usage in general electronic equipment only. This product is not authorized for use in equipment where a higher safety standard and reliability standard is especially required or where a failure of the product is reasonably expected to cause severe personal injury or death, unless the parties have reached an agreement specifying liability governing such use. Würth Elektronik eGmbH & Co. KG products are neither designed nor intended for use in areas such as military, aerospace, aviation, nuclear control, submarine, transportation (aviation, maritime, land, rail, ship control), transportation signal, disaster prevention, medical, public information system etc. Würth Elektronik eGmbH & Co. KG must be informed about the extent of such usage before the design-in stage. In addition, sufficient reliability indicator checks for safety must be performed on every electronic component which is used in electrical circuits that require high safety and reliability functions or performance.

Op dit blad staan de gegevens om dit onderdeel goed te solderen dit is voor fabrikanten die deze componenten gebruiken belangrijk om goeie soldeer connecties te bekomen .

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Packaging Specification - Bulk: [mm]



Size	Packaging Type	Packaging Unit
5 mm x 10 mm x 10 mm	100	100

Würth Elektronik -idm GmbH & Co. KG EBC & Industrial Solutions Max-Eyth-Str. 1 74473 Albstadt Germany Tel: +49 (0) 73 42 945 - 0 www.wa-ele.com info@wa-ele.com		ORDER Date:	ORDER PLS	ORDER ORDER NO. 000 000 000	ORDER ORDER NO. 000 000 000	
WL-TMRW THT Mono-color Round Waterclear		ORDER NO. 151053BS04500				
ORDER NO. 001 010	ORDER NO. 001 010	ORDER NO. 001 010	ORDER NO. 001 010	ORDER NO. 001 010	ORDER NO. 001 010	

This electronic component has been designed and developed for usage in general electronic equipment only. This product is not authorized for use in equipment where a higher safety standard and reliability standard is especially required or where a failure of the product is reasonably expected to cause severe personal injury or death, unless the parties have reached an agreement specifically governing such use. Würth Elektronik -idm GmbH & Co. KG products are neither designed nor intended for use in areas such as military, aerospace, aviation, nuclear control, submarine, transportation (automotive control, train control, ship control, transportation system, disaster prevention, medical, public information system etc.). Würth Elektronik -idm GmbH & Co. KG must be informed about the nature of such usage before the design-in stage. In addition, sufficient reliability evaluation checks for safety must be performed on every electronic component which is used in electrical circuits that require high safety and reliability functions or performance.

Hierin staat hoe de componenten verpakt zijn dit is voor fabrikanten belangrijk in functie van hoe ze de onderdelen dienen aan te bieden aan hun assemblage machines.

PBA EO 1

Kyell De Windt

Mr.mees
Mr.Vereecke**Cautions and Warnings:**

The following conditions apply to all goods within the product series of Optoelectronic Components of Würth Elektronik eiSos GmbH & Co. KG:

General:

- This optoelectronic component is designed and manufactured for use in general electronic equipment.
- Würth Elektronik must be asked for written approval (following the PNP procedure) before incorporating the components into any equipment in fields such as military, aerospace, aviation, nuclear control, submarine, transportation (automotive control, train control, ship control), transportation signal, disaster prevention, medical, public information network, etc. where higher safety and reliability are especially required and/or if there is the possibility of direct damage or human injury.
- Optoelectronic components that will be used in safety-critical or high-reliability applications, should be pre-evaluated by the customer.
- The optoelectronic component is designed and manufactured to be used within the data sheet specified values. If the usage and operation conditions specified in the data sheet are not met, the component may be damaged or discolored.
- Do not drop or impact the components, as the optoelectronic component body may flake apart.
- Würth Elektronik products are qualified according to international standards, which are listed in each product reliability report. Würth Elektronik does not warrant any customer qualified product characteristics beyond Würth Elektronik specifications, for its validity and sustainability over time.
- The responsibility for the applicability of the customer specific products and use in a particular customer design is always within the authority of the customer. All technical specifications for standard products also apply to customer specific products.

Product specific:**Soldering:**

- The solder profile must comply with the Würth Elektronik technical soldering specification. All other profiles will void the warranty.
- All other soldering methods are at the customers' own risk.

Cleaning and Washing:

- Washing agents used during the production to clean the customer application might damage or change the characteristics of the optoelectronic component body, marking or plating. Washing agents may have a negative effect on the long-term functionality of the product.
- Using a brush during the cleaning process may break the optoelectronic component body. Therefore, we do not recommend using a brush during the PCB cleaning process.

Potting:

- If the product is potted in the customer application, the potting material may shrink or expand during and after hardening. Shrinking could lead to an incomplete seal, allowing contaminants into the optoelectronic component body, pins or termination. Expansion could damage the optoelectronic component body, pins or termination. We recommend a manual inspection after potting to avoid these effects.

Storage Conditions:

- A storage of Würth Elektronik products for longer than 12 months is not recommended. Within other effects, the terminals may suffer degradation, resulting in bad solderability. Therefore, all products shall be used within the period of 12 months based on the day of shipment.
- Do not expose the optoelectronic component to direct sunlight.
- The storage conditions in the original packaging are defined according to DIN EN 61760-2.
- For a moisture sensitive component, the storage condition in the original packaging is defined according to PQ/EDEC-J STD-033. It is also recommended to return the optoelectronic component to the original moisture proof bag and reseal the moisture proof bag again.

Handling:

- Violation of the technical product specifications such as exceeding the nominal rated current, will void the warranty.
- The product design may influence the automatic optical inspection.
- Certain optoelectronic component surfaces consist of soft material. Pressure on the top surface has to be handled carefully to prevent negative influence to the function and reliability of the optoelectronic components.
- ESD prevention methods need to be applied for manual handling and processing by machinery.
- Resistors for protection are obligatory.
- Luminaires in operation may harm human vision or skin on a photo-biological level. Therefore direct light impact shall be avoided. All products are additionally certified as risk group 0 to 2 according to DIN EN 62471:2008.
- In addition to optoelectronic components testing, products incorporating these devices have to comply with the safety precautions given in IEC 60625-1 and IEC 62471.

Technical specification:

- The typical and/or calculated values of technical parameters can only reflect statistical figures. The actual parameters of each single product, may differ from the typical and/or calculated values or the typical characteristic line.
- On each reel, only one bin is sorted and taped. The bin is defined on intensity, chromatically coordinate or wavelength and forward voltage.
- In order to ensure highest availability, the reel binning of standard deliveries can vary. A single bin cannot be ordered. Please contact us in advance, if you need a particular bin starting before placing your order to clarify the lead time, MOQ and pricing.

These cautions and warnings comply with the state of the scientific and technical knowledge and are believed to be accurate and reliable. However, no responsibility is assumed for inaccuracies or incompleteness.

Würth Elektronik eiSos GmbH & Co. KG EMC & Inductive Solutions Max-Eyth-Str. 1 74303 Balingen Germany Tel.: +49 (0) 7142 345 - 0 www.w-e-online.com eiSos@we-online.com		ORDER CODE ORDER PLO GENERAL TECHNICAL DATA DIN EN 27660-1en INSTRUCTIONS  
		DESCRIPTION WL-TMRW THT Mono-color Round Waterclear ORDER CODE 151053BS04500
PIN 5 mm	WAVELENGTH 620-630 nm	DATE OF TECHNICAL DATA 2019-11-02 REVISION 01/01 PAGE 1/6

This electronic component has been designed and developed for usage in general electronic equipment only. The product is not authorized for use in equipment where a higher safety standard and reliability is required or where a failure of the product is potentially hazardous to persons, property or the environment (e.g. in aircraft, in medical, in nuclear, in railway, in maritime, in automotive, in industrial, in military, in aerospace, in defense, in police, in fire, in public safety, in transportation, in public information network, etc.). Würth Elektronik eiSos GmbH & Co. KG shall be informed about the nature of such usage before the design-in stage. In addition, sufficient reliability and safety checks for safety must be performed in every relevant component which is used in a restricted circuit that requires high safety and reliability, before its application.

Important Notes

The following conditions apply to all goods within the product range of Würth Elektronik eiSos GmbH & Co. KG:

1. General Customer Responsibility

Some goods within the product range of Würth Elektronik eiSos GmbH & Co. KG contain statements regarding general suitability for certain application areas. These statements about suitability are based on our knowledge and experience of typical requirements concerning the areas, serve as general guidance and cannot be estimated as binding statements about the suitability for a customer application. The responsibility for the applicability and use in a particular customer design is always safely within the authority of the customer. Due to this fact it is up to the customer to evaluate, where appropriate to investigate and decide whether the device with the specific product characteristics described in the product specification is valid and suitable for the respective customer application or not.

2. Customer Responsibility related to Specific, in particular Safety-Related Applications

It has to be clearly pointed out that the possibility of a malfunction of electronic components or failure before the end of the usual lifetime cannot be completely eliminated in the current state of the art, even if the products are operated within the range of the specifications. In certain customer applications requiring a very high level of safety and especially in customer applications in which the malfunction or failure of an electronic component could endanger human life or health it must be ensured by most advanced technological aid of suitable design of the customer application that no injury or damage is caused to third parties in the event of malfunction or failure of an electronic component. Therefore, customer is cautioned to verify that data sheets are current before placing orders. The current data sheets can be downloaded at www.w-e-online.com.

3. Best Care and Attention

Any product-specific notes, cautions and warnings must be strictly observed. Any disregard will result in the loss of warranty.

4. Customer Support for Product Specifications

Some products within the product range may contain substances which are subject to restrictions in certain jurisdictions in order to serve specific technical requirements. Necessary information is available on request. In this case the field sales engineer or the internal sales person in charge should be contacted who will be happy to support in this matter.

5. Product R&D

Due to constant product improvement product specifications may change from time to time. As a standard reporting procedure of the Product Change Notification (PCN) according to the JEDEC-Standard inform about minor and major changes. In case of further queries regarding the PCN, the field sales engineer or the internal sales person in charge should be contacted. The basic responsibility of the customer as per Section 1 and 2 remains unaffected.

6. Product Life Cycle

Due to technical progress and economical evaluation we also reserve the right to discontinue production and delivery of products. As a standard reporting procedure of the Product Termination Notification (PTN) according to the JEDEC-Standard we will inform at an early stage about inevitable product discontinuance. According to this we cannot guarantee that all products within our product range will always be available. Therefore it needs to be verified with the field sales engineer or the internal sales person in charge about the current product availability expectancy before or when the product for application design-in disposal is considered. The approach named above does not apply in the case of individual agreements deviating from the foregoing for customer-specific products.

7. Property Rights

All the rights for contractual products produced by Würth Elektronik eiSos GmbH & Co. KG on the basis of ideas, development contracts as well as models or templates that are subject to copyright, patent or commercial protection supplied to the customer will remain with Würth Elektronik eiSos GmbH & Co. KG. Würth Elektronik eiSos GmbH & Co. KG does not warrant or represent that any license, either expressed or implied, is granted under any patent right, copyright, mask work right, or other intellectual property right relating to any combination, application, or process in which Würth Elektronik eiSos GmbH & Co. KG components or services are used.

8. General Terms and Conditions

Unless otherwise agreed in individual contracts, all orders are subject to the current version of the "General Terms and Conditions of Würth Elektronik eiSos Group", last version available at www.w-e-online.com.

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Op deze pagina's staat vooral juridische gegevens die verplicht zijn om de fabrikant in te dekken bij eventuele verkeerd gebruik van de componenten en of rechtszaken.

Methode's om diode-karakteristiek te meten:

Het meten van een diode-karakteristiek kan op verschillende manieren worden uitgevoerd, afhankelijk van de beschikbare apparatuur en het gewenste detailniveau.

Hier zijn twee methoden:

1. Handmatige meting met een voedingsbron en multimeter

Deze methode is eenvoudig en maakt gebruik van basisonderdelen.

Benodigdheden:

- Variabele gelijkspanningsvoeding
- Multimeter
- Weerstand (om de stroom te beperken)
- De diode die je wilt testen

Stappen:

1. Circuit opbouwen:

- Verbind de voedingsbron in serie met een stroombeperkende weerstand en de diode.
- Sluit de multimeter parallel aan de diode om de spanning over de diode te meten.

2. spanning verhogen:

- Stel de voedingsspanning stap voor stap in (bijvoorbeeld van 0 V naar een hogere waarde).
- Meet bij elke stap de spanning over de diode (forward voltage) en de stroom door de diode.

3. Gegevens noteren:

- Noteer de spanningen en stromen en plot de gegevens om de diode-karakteristiek te visualiseren.

2. Meting met een curve tracer of functiegenerator

Deze methode is sneller en levert een directe visualisatie van de karakteristiek.

Benodigdheden:

- Curve tracer of oscilloscoop met functiegenerator
- Meetprobes
- Diode

Stappen:

1. Aansluiten:

- Verbind de diode met de curve tracer of sluit deze aan op de functiegenerator en oscilloscoop.
- Zorg ervoor dat de meetinstrumenten ingesteld zijn op een geschikte frequentie en amplitude.

2. Signaal genereren:

- Gebruik een sinus- of triangulaire golfvorm om spanning over de diode te variëren.
- De curve tracer of oscilloscoop zal de stroom door en de spanning over de diode in real-time plotten.

3. Aflezen:

- Lees direct de karakteristieke curve af van het scherm van de curve tracer of oscilloscoop.

Beide methoden zijn nuttig, maar de keuze hangt af van de gewenste precisie en de beschikbare apparatuur. Voor een snelle meting of analyse wordt een curve tracer aanbevolen.

Ik zou het liefst werken met methode 1 omdat men daar de diode beschermd door middel van een weerstand en men hier langzaam aan de grensen kan opzoeken .

Met methode 2 als je een fout maakt is het direct defect + je hebt ook duurdere toestellen nodig.

We voorzien een voedingspanning van 5 V daar dit een zeer gebruikelijke spanning is in electronica.

Ik heb getest tot 6V om zo echt de grenzen te kunnen opzoeken.

Ik heb mijn bron ook ingesteld op 1A dit omdat we ons wat ruimte willen geven i.v.m. het opzoeken van de grensen als ik deze zou instellen op 0.020A om de diode te beschermen zal ik nooit de limieten van de diode kunnen vinden.

Zoals hierboven vermeld moete we zeker de stroom door de led en weerstand in het oog houden te hoge stromen kunnen de componenten kapot maken.

Tevens moet men ook opletten welke spanningen we aansluiten omdat de diode in verkeerd aangesloten configuratie maar 5V kan verdragen.

Het vermogen is evenredig met de stroom en deze mag ook zeker niet overschreden worden.

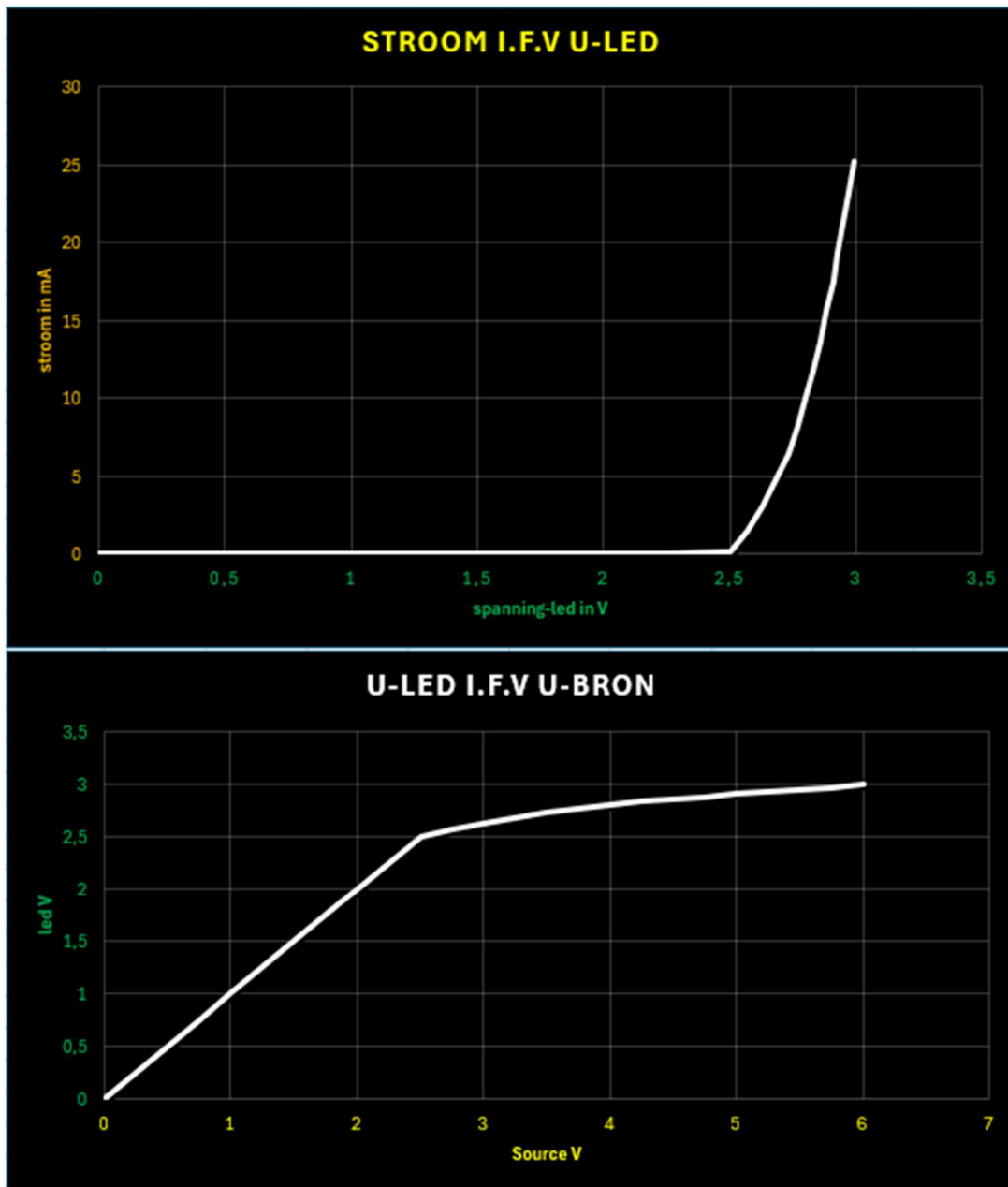
Opmeten van de waarden en tekenen van de karakteristiek

Ubron	Uled	I = mA
0	0	0
0,25	0,25	0,0006
0,5	0,5	0,0006
0,75	0,75	0,0005
1	1	0,0007
1,25	1,25	0,0006
1,5	1,5	0,0007
1,75	1,75	0,0005
2	2	0,0004
2,25	2,25	0,0007
2,5	2,5	0,236
2,75	2,57	1,495
3	2,63	3,05
3,25	2,68	4,7
3,5	2,73	6,43
3,75	2,77	8,2
4	2,8	10
4,25	2,83	11,8
4,5	2,86	13,7
4,75	2,88	15,56
5	2,91	17,47
5,25	2,93	19,38
5,5	2,95	21,3
5,75	2,97	23,24
6	2,99	25,2

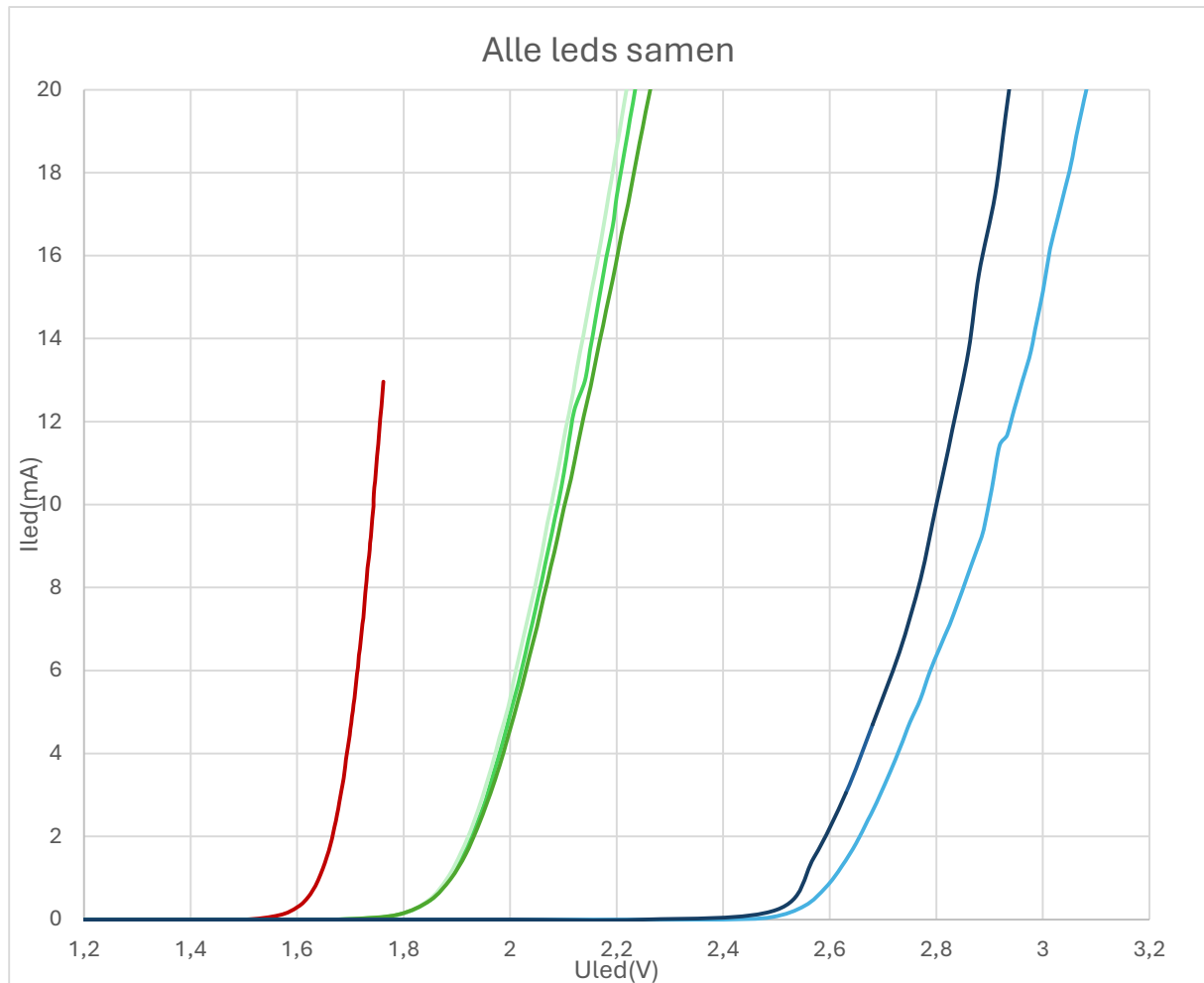
PBA EO 1

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Reflectie



Het verschil is te verklaren door volgende redenen.

Elk kleur heeft zijn eigen materialen en eigenschappen omdat elk kleur ook ergens anders in het spectrum zit.

Ook de test omgevingen en gebruikte toestellen heb een effect op de resultaten .

Je kan hier ook zien dat de gelijkaardige kleuren dicht bij elkaar zitten .